

Comments

- Only focus on compressing the covariance matrix and work with this, not supervised and unsupervised learning
- Set up quickly to test

Second-Order Pooling for Protein Language Models

Protein Prediction I for Bioinformaticians



Joel Simon



Julius Schmidt



Lisa Börner



Andreas Weitz

The Status Quo: Mean Pooling

Our Data

- **Input:** pLM per residue information
- **Output:** per protein information
- **For what:** analyzing predicting protein properties

Mean Pooling

- Mean over all residues
- Status quo
- destroys co-occurrence information

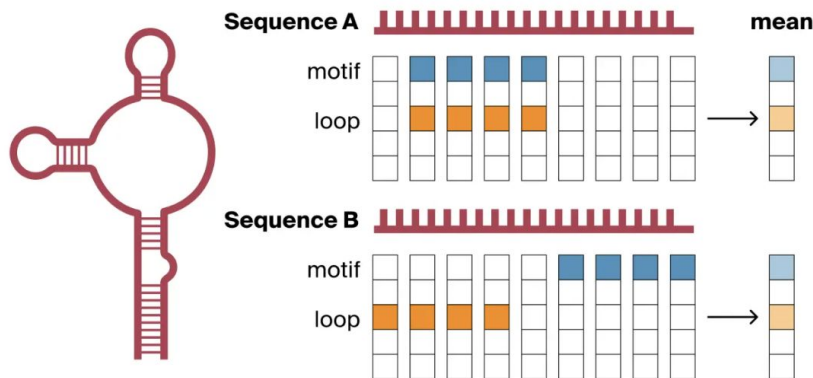


Fig.1: Can't tell if motif and loop occur at the same time

Research Question

Does covariance-based pooling of per-residue ProtX embeddings improve per-protein downstream task performance compared to mean pooling, and at what embedding size does it become parameter-efficient?

Motivation: covariance pooling

Alternative: Covariance Pooling

- Uses the **second moment** of residue activations instead of the first
- Captures how features **co-activate** across positions
- Preserves combinatorial structure
- Reports **substantial gains in genomic language models (gLMs)**
- **Unexplored in protein language models (pLMs)**

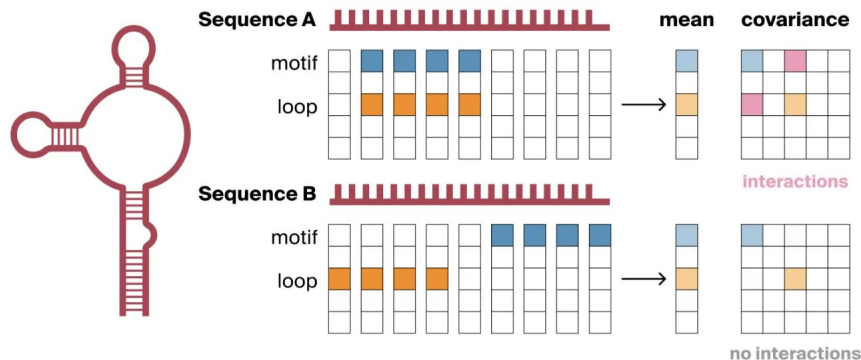
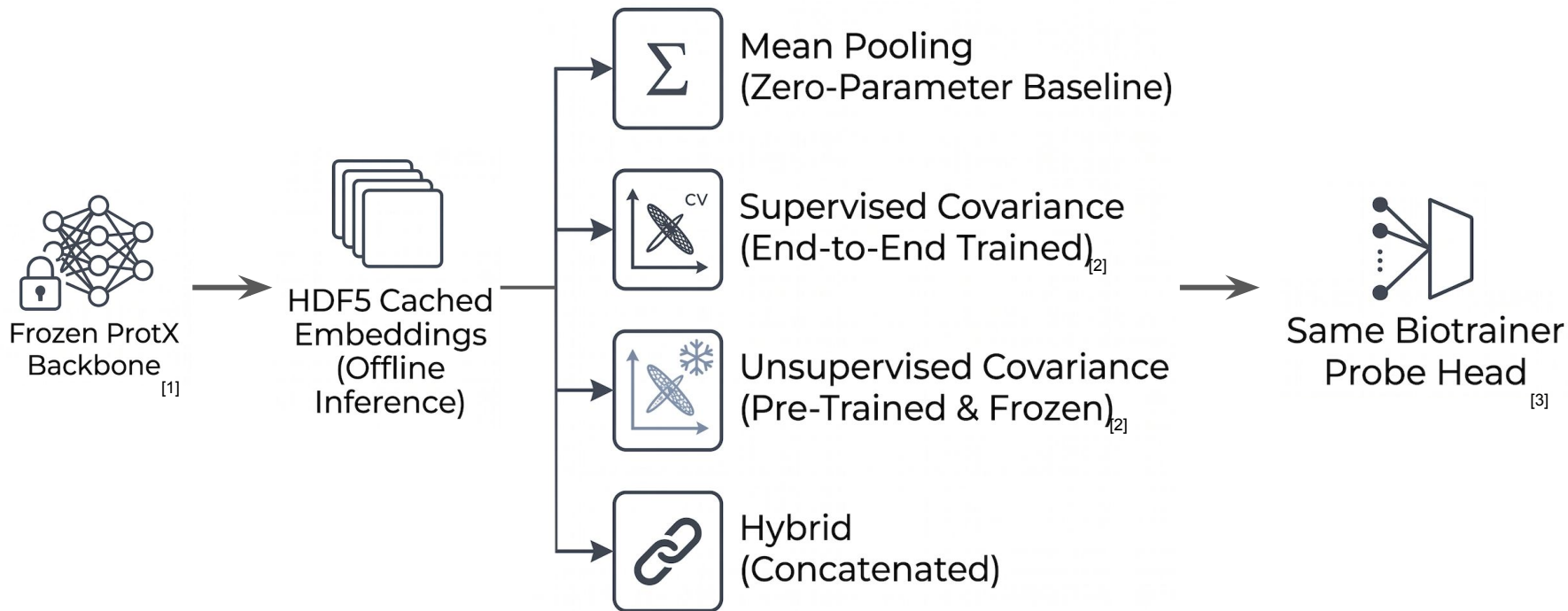


Fig.2: comparison mean pooling and covariance pooling on two sequences with different properties

Experimental setup



Supervised vs. Unsupervised

Supervised Learning

- Saves important information in covariance matrix
- Adjusting the weights, highlighting the important information and neglecting other
- Must retrain for every task

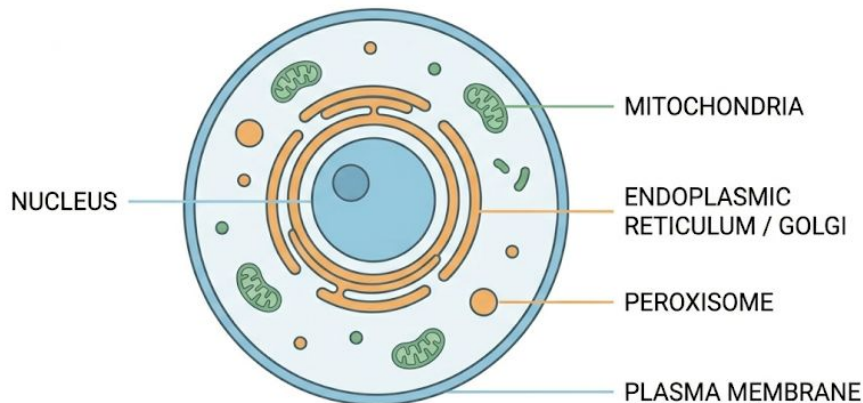
Unsupervised Learning

- Saves all information in covariance matrix
- Less precision due to compression
- Train once for all tasks

Experimental setup - Tasks

Subcellular Localization

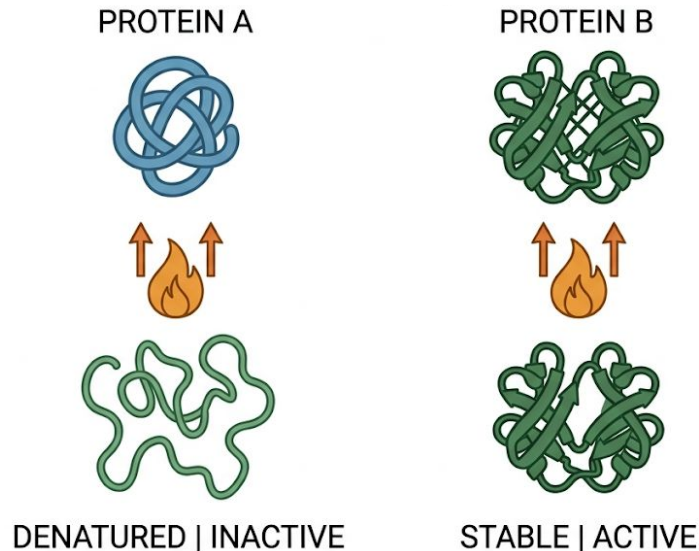
- **Objective:**
10-class classification predicting a protein's cellular compartment.
- **Protocol:**
Uses residues_to_class to pool^[3] per-residue embeddings.
- **Datasets:**
Evaluated on DeepLoc^[4] and FLIP benchmarks.
- **Metric:**
Accuracy / Q10 (measures 10-state prediction accuracy).



Experimental setup - Tasks

Meltome (Thermostability)

- **Objective:**
Regression task predicting continuous melting temperatures.
- **Protocol:**
Utilizes the residues_to_value protocol.^[4]
- **Dataset:**
Exclusively uses the FLIP Meltome dataset.^[4]
- **Metric:**
Spearman R (measures rank correlation between predicted and true values).



Experiments : Evaluating Pooling Methods



Core Experiments

- **Head-to-head benchmark:**
4 pooling methods * 2 tasks * 3 seeds = 24 main results.
- **Size-matched efficiency sweep:**
Vary projection dimension in 8, 16, 24, 32, 48.
- **Layer sweep:**
Repeat benchmarks across different layers (last, middle, early).
- **Stretch Goals (if core finishes early):**
Attention pooling baselines.

Next steps:

Environment & Backbone Setup

- Set up environment with biotrainer and ProtX.:
- Extract last-layer per-residue embeddings for DeepLoc and Meltome datasets.
- Start implementation

1 Backbone

- **Model:** ProtX (provided), frozen.
- **Layer:** last-layer per-residue embeddings.
- **Caching:** extract embeddings once using the provided inference script; save to disk (HDF5). All subsequent experiments run on cached tensors.



Backup Slides

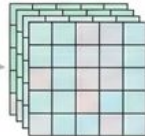


Input Protein Sequence
[Length L, Dimension d]

A C N A I V B K N P



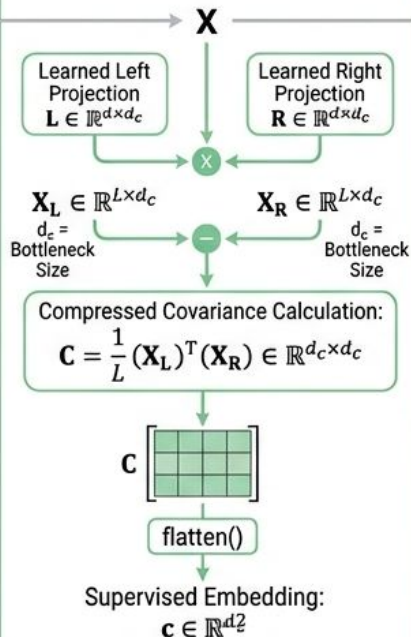
3: Per-Residue
Embeddings
(Pre-Pooling):



Per-Residue
Embeddings
(Pre-Pooling):
 $\mathbf{X} \in \mathbb{R}^{L \times d}$

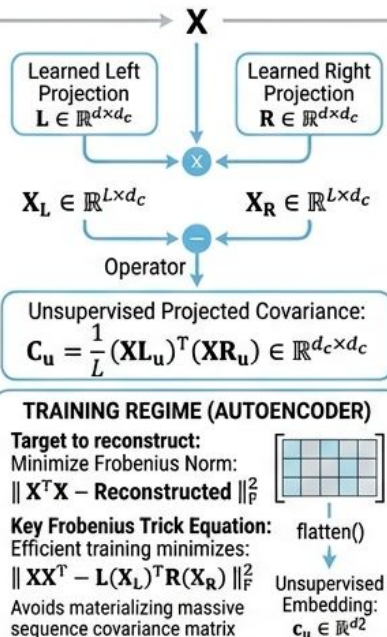
Masking & Padding
applied for pooling

SUPERVISED COVARIANCE POOLING (Task-specific)



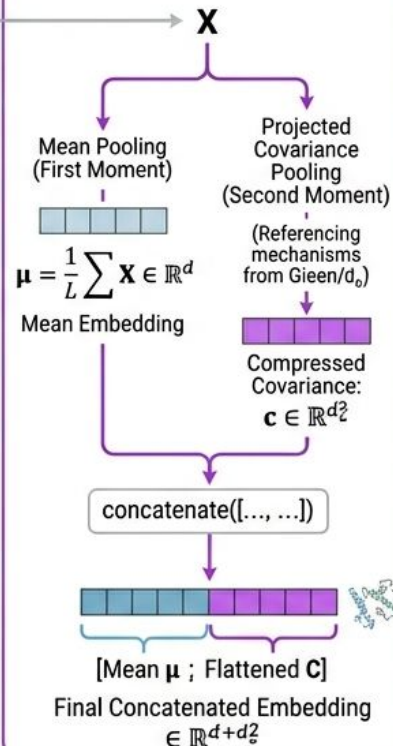
Regime: L and R are optimized end-to-end with the downstream probe head for task performance

UNSUPERVISED COVARIANCE POOLING (Task-agnostic)

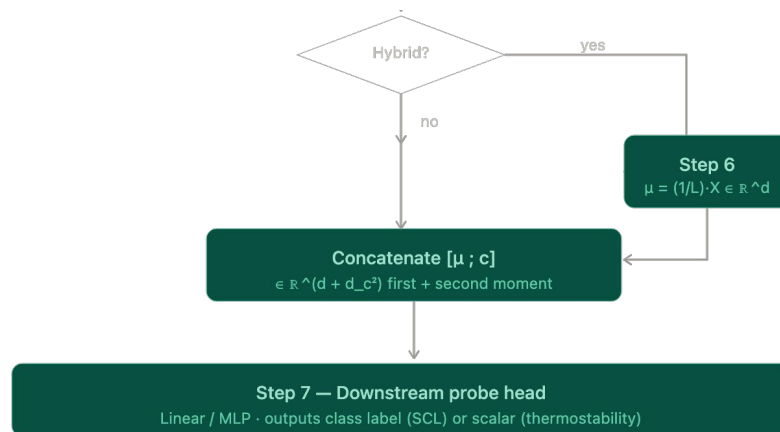
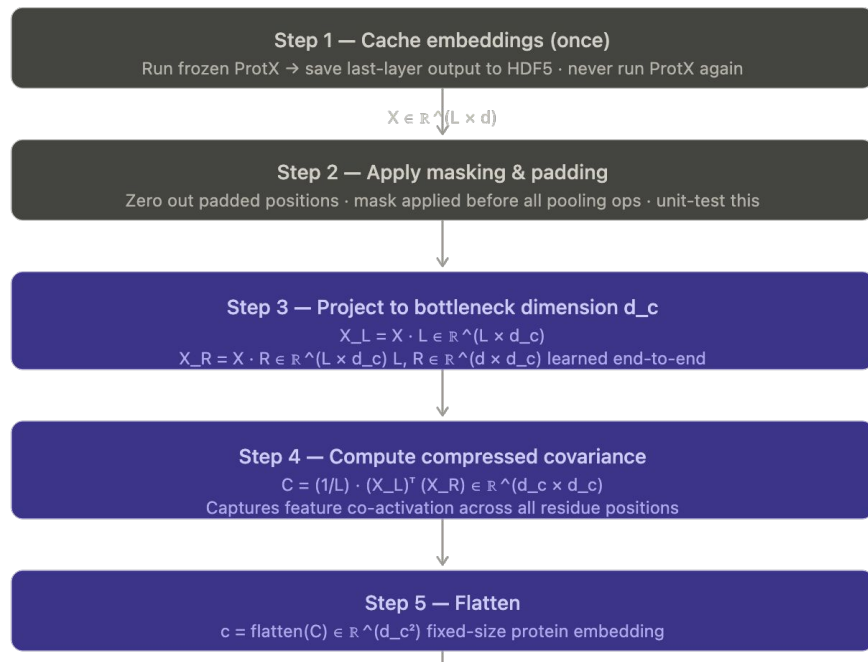


Regime: $\mathbf{L}_w$, $\mathbf{R}_u$ are pre-trained to reconstruct the full covariance and then frozen for all downstream tasks

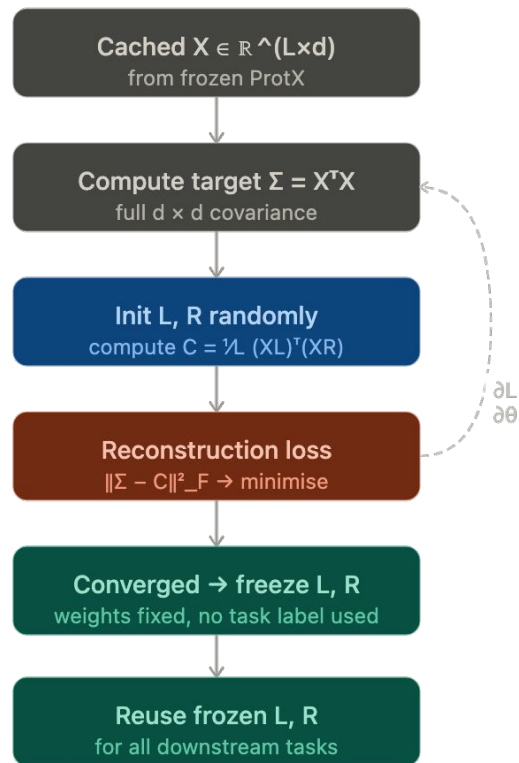
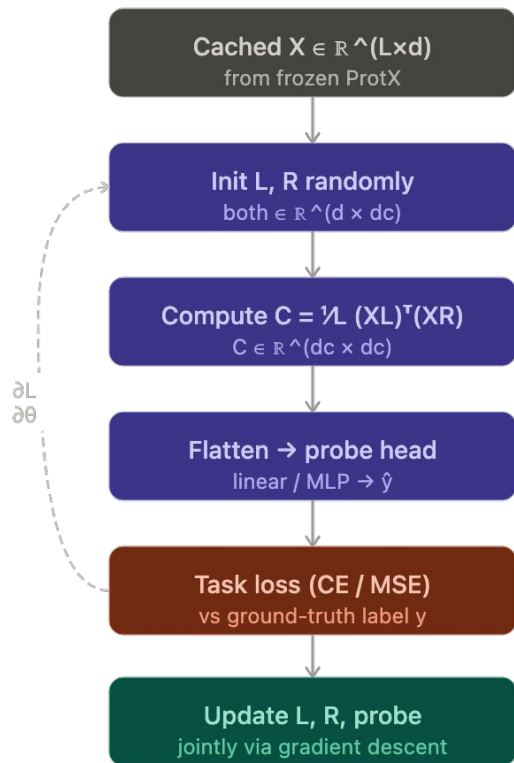
HYBRID (CONCATENATED) POOLING



These detailed embeddings are generated once per protein for use by subsequent probes.



ProtX backbone remains frozen throughout · only L, R, and probe are trained



Visualization Plan



- **Main results:**
Bar charts per task with error bars across the 3 random seeds.
- **Efficiency curve:**
Downstream metric vs. embedding size.
- **Matrix Visualization:**
Show learned covariance embeddings as heatmaps for representative proteins.
- **Clustering:**
t-SNE/UMAP projections comparing flattened covariance vs. mean-pooled embeddings by class label.